Place Value

Exam code: 0607

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Place Value

What is place value?

- When a number is written down using digits, each digit has a **value** depending on its position **(place)** within the number
- Each place has a value **ten times** larger than the place to the right of it
- e.g. For the number 9876
  - The 6 represents 6 **ones** (or units) (6)
  - The 7 represents 7 **tens** (70)
    - "ten" is ten times larger than "one"
  - The 8 represents 8 **hundreds** (800)
    - "hundred" is ten times larger than "ten"
  - The 9 represents 9 **thousands** (9000)
    - "thousand" is ten times larger than "hundred"
  - In words, this number is **nine thousand, eight hundred and seventy six**

How do I read large numbers?

- Start with the ones (units) digit and work 'right to left' through the digits to deduce the place value that the number starts with
  - e.g. For the number 12345678

Ten Millions	Millions	Hundred Thousands	Ten Thousands	Thousands	Hu
1	2	3	4	5	

- 12 345 678 starts in the ten millions place value
- So it would be read (and written in words) as **twelve million, three hundred and forty-five thousand, six hundred and seventy-eight**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Examiner Tips and Tricks

For example, 54687321 is easier to read as 54 687 321.

Large numbers are normally written with spaces between the groups of three digits.

How does place value work for decimals?

- Starting with the decimal point
  - digits to the **left** of the decimal point form the **whole** number part (ones, tens, thousands, ...)
  - digits to the **right** of the decimal point form the **decimal** part
- Each decimal place has a value **ten times** larger than the place to the right of it
- e.g. For the number 36.952
  - The whole number part is 36 (3 tens and 6 ones)
  - The 9 represents 9 **tenths** (0.9)
    - "one" is ten times larger than "tenth"
  - The 5 represents 5 **hundredths** (0.05)
    - "tenth" is ten times larger than "hundredth"
  - The 2 represents 2 **thousandths** (0.02)
    - "hundredth" is ten times larger than "thousandth"
  - In words, this number is **thirty six point nine five two**

How do I read decimals?

- The whole number part would be read as above
- The decimal part is read digit by digit
  - e.g. The number 23.45678 would be read (and written in words) as **twenty three point four five six seven eight**
- Although they are not read, it is still important to know the value of each decimal place

Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths
2	3	.	4	5	6

- You will often hear these place values used relating to race time
  - e.g. In a sprint race, athletes may be separated by "five hundredths of a second" (0.05 seconds)

Worked Example

Answer:

Note down the value of each digit

Ten Thousands	Thousands	Hundreds	Tens	Ones
8	7	6	5	4

7 000 or seven thousand

(b) A racing car completed a lap of a circuit in 1 minute and 14.263 seconds. Write down the value of the digit 3.

Answer:

Note down the value of each digit, starting with the decimal point
Work to the left (of the decimal point) for the whole number part (14)
Work to the right (of the decimal point) for the decimal part (263)

Tens	Ones	Point	Tenths	Hundredths	Thousandths
1	4	.	2	6	3

0.003 seconds or 3 thousandths of a second

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